Cardinality of Crisp Sets vs. Fuzzy Sets: The Concept of Counting

PresentedBy:SK ENAMUL RAHAMAN

Name :SK ENAMUL RAHAMAN

University Roll No : 12230623048

Registration No :231220110131 0f 2023

Semester: 6thSem

Subject: Soft Computing(PCCAIML603 & PCCAIML693)

Department of Artificial Intelligence and Machine Learning

St. Thomas' College of Engineering and Technology

Set Cardinality in Soft Computing

- Introduction to Set Cardinality
- Cardinality of Crisp Sets
- Cardinality of Fuzzy Sets
- Mathematical Formulation
- Comparison: Crisp vs Fuzzy Cardinality
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Introduction (1/2)

Cardinality

refers to the **count or size of a set**

In classical (crisp) sets

elements either **belong or do not belong**

In fuzzy sets

elements have **degrees of membership**

Hence

counting in fuzzy sets is **not straightforward**

Introduction (2/2)

- Soft Computing deals with **uncertainty and imprecision**
- Fuzzy logic extends classical set theory
- Cardinality concept must be **redefined** for fuzzy sets
- This leads to the idea of **fuzzy cardinality**

Fuzzy Set

vs.

Crisp Set

Cardinality of Crisp Sets

- Crisp set: elements have **binary membership**
- Membership function: *\mu(x)\in\{0,1\}*
- Cardinality is the **total number of elements**

Example: If *A=\{2,4,6,8\}* Then,

$$X_A(x) = \begin{cases} 1 \text{ if } x \in A \\ 0 \text{ if } x \notin A \end{cases}$$

Cardinality of Fuzzy Sets

- Fuzzy set allows **partial membership**
- Membership function: *\mu(x)\in[0,1]*
- Cardinality is defined as the **sum of membership values**

Example: If

$$A = \{(x_1, 0.3), (x_2, 0.7), (x_3, 1.0)\}$$

Then,

$$|A| = 0.3 + 0.7 + 1.0 = 2.0$$

Mathematical Formulation

Crisp Set Cardinality:

$$|A| = \sum \mu_A(x) \quad \text{where } \mu_A(x) \in \{0, 1\}$$

Fuzzy Set Cardinality:

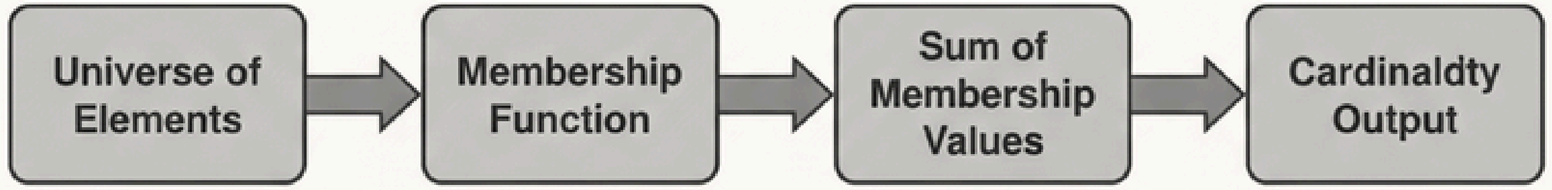
$$|A| = \sum_{x \in X} \mu_A(x) \quad \text{where } \mu_A(x) \in [0, 1]$$

- Result may be a **real number**

Comparison: Crisp vs Fuzzy Cardinality

Feature	Crisp Set	Fuzzy Set
Membership	Binary (0 or 1)	Continuous (0–1)
Cardinality	Integer	Real number
Uncertainty	Not allowed	Allowed
Flexibility	Rigid	Flexible

Conceptual Block Diagram



A visual represetation of set counting using fuzzy logic principles.

- Diagram shows **how fuzzy counting is performed**
- Emphasizes **soft decision-making**

Applications in Soft Computing

- Pattern recognition
- Decision-making systems
- Medical diagnosis
- Image processing
- Artificial Intelligence systems
- Linguistic variable analysis

References / Bibliography (IEEE Format)

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